Does Anatomy-Guided Masking Help? A Controlled Comparison of Six Masking Policies for Joint-Embedding Predictive Pretraining on Retinal OCT

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Abstract

Masked predictive pretraining asks a model to reconstruct the representation of hidden image regions. Which regions are hidden is a free design choice, and in medical imaging there is an intuitive answer: hide the tissue that carries the diagnosis. We test that intuition under tightly controlled conditions. Six masking policies are continued from a single shared I-JEPA checkpoint on 3D retinal OCT, differing in how predictor targets are placed, and evaluated with a frozen linear-probe protocol for glaucoma classification (FairVision, N=3000). Each policy was continued once, so what follows describes these runs rather than an expected ranking over retrainings. Guidance helped: restricting targets to retinal tissue improves AUC by $+0.0120$ over unguided masking at the matched epoch (95% CI $[+0.0068, +0.0173]$), and a band located by a first-order intensity statistic improves it by $+0.0109$ at epoch 100 ($[+0.0057, +0.0160]$, p<0.0001). All six contrasts against the null, at all three epochs, exclude zero. But anatomical *precision* does not reliably help. A policy that places 97% of masked patches on tissue, using a trained retinal segmentation model to shape the targets, matches unguided masking at epoch 50 ($+0.0013$) and falls *below* it by epoch 75 (-0.0111, CI $[-0.0181, -0.0042]$), as does a coverage-constrained policy ($+0.0002$ at epoch 50) — while the two policies that merely *place* rectangles on tissue gain an order of magnitude more. Carried to epoch 100 the coverage-constrained policy does not merely stall, it peaks at epoch 73 and then declines, ending -0.0168 below unguided masking (CI $[-0.0233, -0.0105]$); an audit found its targets are shortened after placement, so it does not test aggressive coverage. Direct measurement of mask geometry across 600 slices shows the most anatomically precise policy is not the best performer. The best policy consults no segmentation model at all. A subgroup audit over 23 probes finds the same worst-served group every time; every disease stage improves with intervals excluding zero (largest for mild, $+0.0137$, though the gains are not separated from one another), while among race groups only the white subgroup's gain excludes zero. The difficulty ordering is untouched: the severe-to-mild gap stays within 0.0114 of constant across every policy.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

1 Introduction

Self-supervised learning has become the default route to representation quality in medical imaging, where expert annotation is expensive and pathology is rare [Azizi et al., 2021, Zhou et al., 2023, Qiu et al., 2024]. Joint-embedding predictive architectures such as I-JEPA [Assran et al., 2023] learn by predicting the *representation* of masked target blocks from a visible context block, avoiding pixel reconstruction and its bias toward high-frequency detail [He et al., 2022, Xie et al., 2022].

I-JEPA samples its targets geometrically: rectangles placed uniformly at random. In natural images this is defensible, since informative content is spread broadly across the frame. In a retinal OCT B-scan it is not obviously so. A large fraction of the image is dark vitreous and background; the diagnostic signal for glaucoma is concentrated in a thin band of retinal tissue. This motivates an intuitive hypothesis, one that several recent methods adopt [Li et al., 2022, 2024, Wang et al., 2025, Shi et al., 2022]:

> *Direct the predictive objective at clinically meaningful anatomy rather than at arbitrary image regions.*

We test this hypothesis directly and at several strengths. Rather than proposing one anatomy-aware method and reporting that it beats a baseline, we construct a *ladder* of masking policies of increasing anatomical specificity — from unguided rectangles, through rectangles aimed at tissue, to ragged targets whose shape follows the segmented retina — and hold everything else fixed. All arms continue from one shared pretrained checkpoint, use identical optimiser, schedule, and effective batch size, and are evaluated under one frozen-probe protocol.

Guidance helps, but the benefit does not increase with anatomical precision. The two policies whose targets track the segmented retina most faithfully are statistically indistinguishable from unguided masking at epoch 50, and the one we carried further falls significantly below it by epoch 75, while two policies that merely place ordinary rectangles on tissue gain an order of magnitude more. One earlier shaped policy did exceed its rectangle counterpart at epoch 30, so the anatomy result is mixed rather than uniformly negative. The best performer locates a band by a per-column intensity centroid.

Contributions.

- A controlled six-arm study isolating masking policy in medical SSL, with a shared ancestor checkpoint and a single evaluation protocol (every probe fitted is listed in Appendix A).
- Evidence that in these runs guidance helps while anatomical precision does not reliably add to it, including the specific control — rectangles aimed at retinal tissue — that separates “avoid background” from “target anatomy”.
- Direct measurement of mask geometry for every policy on 600 slices, showing that the most anatomically precise policy is not the best performer, and an explicit accounting of the axes on which the anatomy arms are confounded.
- A subgroup audit and a 10^{-5} reproduction of the headline result from public weights on different hardware and precision.

2 Related work

Masked image modelling and JEPAs. Masked autoencoders reconstruct pixels [He et al., 2022, Xie et al., 2022]; BEiT [Bao et al., 2022] and iBOT [Zhou et al., 2022] predict discrete tokens; data2vec [Baevski et al., 2022] and I-JEPA [Assran et al., 2023] predict latent representations. Video and 3D extensions follow the same template [Bardes et al., 2024, Chen et al., 2023b, Taleb et al., 2020]. Recent analyses question how much of the benefit derives from the masking distribution itself [Balestriero and LeCun, 2025, He et al., 2025].

Informed masking. A substantial line of work argues that *what* is masked matters. ADIOS [Shi et al., 2022] learns an adversarial masking model; SemMAE [Li et al., 2022] uses learned semantic parts; AttMask [Kakogeorgiou et al., 2022] masks by attention; HardPatches [Wang et al., 2023a] and AutoMAE [Chen et al., 2023a] target difficulty. In the medical domain, AnatoMask [Li et al., 2024], AnatomyMAE [Ceballos Arroyo et al., 2026], MSMAE [Mao et al., 2023], and VasoMIM [Huang et al., 2026] explicitly steer masking toward anatomy, and Wang et al. [2025] argue directly for masking clinically relevant regions. Our study is a controlled test of that family's central premise on one task.

Evidence that informed masking is not uniformly better. The picture is less one-sided than that suggests. The original masked-modelling papers report that random sampling is hard to beat: MAE's ablation puts random-75 at 73.5 linear against block-75 at 63.9 [He et al., 2022], and SimMIM finds random beats its best block variant [Xie et al., 2022]. Among informed schemes, AttMask sits within 0.2 points of random [Kakogeorgiou et al., 2022], AutoMAE ties MAE under full fine-tuning [Chen et al., 2023a], HPM gains only when randomness is retained and *loses* when restricted to the hardest patches [Wang et al., 2023b], and attention-guided MAE underperforms MAE in the low-label regime [Sick et al., 2025]. Most directly, SemMAE's own part-quality ablation shows that *imperfect* semantic parts add nothing over no parts at all [Li et al., 2022], and Guo et al. [2025] reproduce SemMAE below random MAE under a common protocol, with the deficit largest under frozen evaluation — also our protocol. Chen et al. [2023a] name the mechanism a "patch selection dilemma": a slight foreground bias helps, but raising it too far starves the encoder of the evidence it must predict from. Conversely, the closest positive precedent for our best policy is segmentation-free too: Lee et al. [2025] select foreground from normalised intensity alone and gain across five 3D medical benchmarks. Appendix K tabulates these.

That random masking is a strong baseline, and semantic guidance wins only sometimes, is known and contested; we do not overturn that literature. We add a controlled measurement of where a *trained medical segmentation model* falls on that spectrum, in a setting — 3D retinal OCT, joint-embedding prediction, frozen probe — where we could find no prior controlled comparison.

Retinal foundation models, OCT, and fairness. RETFound [Zhou et al., 2023], VisionFM [Qiu et al., 2024], URFound [Yu et al., 2024], and OCTCube [Liu et al., 2026] pretrain on large retinal corpora; OCT-specific masked modelling is explored by Pissas et al. [2024]. MIRAGE [Morano et al., 2025] provides multimodal OCT segmentation and supplies the anatomical guidance used by several of our arms. Deep glaucoma detection from OCT is established [Maetschke et al., 2019, Ran et al., 2019, Noury et al., 2022]. Performance disparities across demographic groups are widely documented [Seyyed-Kalantari et al., 2021, Larrazabal et al., 2020, Gichoya et al., 2022]. FairVision [Luo et al., 2024a] and Harvard-GF [Luo et al., 2024b] provide OCT data with demographic attributes; FairMedFM [Jin et al., 2024] and Shi et al. [2025] benchmark fairness for medical foundation models.

3 Method: a ladder of masking policies

3.1 Background

I-JEPA partitions the token grid of an image into a context block and M target blocks. A context encoder f_θ embeds the visible tokens; a predictor g_ϕ maps that embedding plus positional queries to predicted representations of the target tokens; an exponential-moving-average target encoder $f_{\bar{\theta}}$ supplies the regression targets. With a 16×16 token grid the design choice we vary is which of the 256 cells become targets.

3.2 The six policies

All policies produce $M=4$ target blocks and one context block under identical scale and aspect-ratio ranges. The rectangle arms differ only in placement.

RANDOM (null). Stock I-JEPA multiblock. Four rectangles placed uniformly at random anywhere in the grid. No image content is consulted.

ENVELOPE (random-within-retina). Identical rectangles — same shape, size and count as RANDOM — but rejection-sampled onto the retinal envelope predicted by MIRAGE [Morano et al., 2025]. Only *location* changes. This is the control that separates "stop wasting targets on background" from "target anatomy specifically".

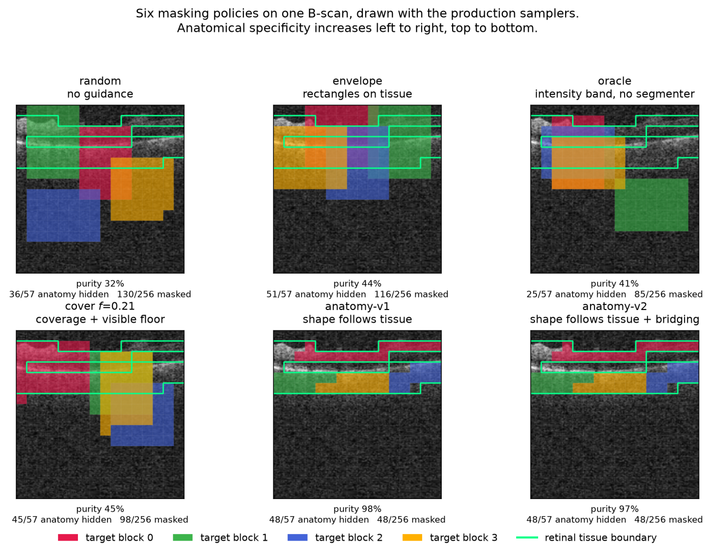

Figure 1: The six masking policies on one identical B-scan, drawn with the production samplers. Each predictor target block has its own colour; the green contour is the anatomy reference. Anatomical specificity increases along the ladder: ENVELOPE and COVER place rectangles on tissue, while the anatomy arms grow targets that trace the retina almost exactly. Downstream AUC does not increase with that specificity (Table 1). Figure 4 repeats this over five B-scans spanning the volume.

CENTROID (segmentation-free). A band whose vertical position is located per slice by a per-column intensity-weighted row centroid, smoothed across columns. It adapts to the retina's position and curvature from a first-order intensity statistic, with no segmentation model and no ground-truth annotation of any kind.

ANATOMY-V1 / ANATOMY-V2. Ragged connected components grown on the MIRAGE score map, so target *shape* follows tissue rather than being rectangular. v2 additionally bridges diagonal adjacencies.

COVER f. Targets chosen greedily to cover the anatomy subject to a hard floor leaving a fraction f of tissue visible in the context. We pretrain $f{=}0.21$. An implementation defect means the delivered masks never reach f (Appendix E).

Figure 1 shows all six on the same B-scans.

3.3 Hypotheses

The ladder yields three testable statements:

H1 *Avoiding background helps.* ENVELOPE $>$ RANDOM.

H2 *Anatomical precision helps further.* anatomy arms $>$ ENVELOPE.

H3 *Any benefit is due to anatomical targeting, not to incidental changes in mask area, target count, or context size.*

4 Experimental setup

Data. FairVision [Luo et al., 2024a] glaucoma OCT volumes. Each volume is a 3D scan resampled to 100 B-scans of $256{\times}256$ after transform. The label is a binary glaucoma diagnosis. Splits are fixed and patient-disjoint as released; downstream evaluation uses a held-out test split of $N{=}3000$ volumes (1466 positive / 1534 negative), seed 42, fixed loader order. No test volume is seen during pretraining, probe fitting, or model selection: the probe head is selected on a separate validation split. Labels are byte-identical across arms, so any two arms are paired-comparable on the same cases. The dataset carries self-reported demographic attributes, which we use only for the subgroup analysis in Section 5.4 and never as model inputs.

Pretraining. Patch-level I-JEPA, ViT-Base, 256×256 crops, patch size 16, predictor depth 6, EMA target encoder. **Every arm resumes from the same epoch-25 checkpoint** and continues with identical optimiser, learning-rate and weight-decay schedules, and effective batch size (512 for all arms). Guidance is ramped linearly from epoch 25 to 30 and held at full strength thereafter. Within the rectangle family the *only* difference between arms is the mask sampler (Section 3); the anatomy family additionally differs in collation, and we quantify that in Section 5.

Arms differ in how far they were carried. RANDOM, CENTROID and ENVELOPE reach epoch 100. ANATOMY-V2 reaches epoch 75 (its epoch-75 and epoch-92 probes under an earlier target-precision splice are excluded and replaced by a clean fp32 continuation, Appendix D); we stopped that arm having seen its epoch-75 deficit, which is a selective stopping horizon and a real weakness — a trajectory can recover, and its epoch-100 value is unmeasured, not known to be worse. ANATOMY-V1 has only epoch 30. COVER $f=0.21$ reaches epoch 100, with an extra probe at its epoch-73 peak. Epoch 50 is therefore the only epoch at which five policies are simultaneously available, and it carries every cross-policy comparison that involves the anatomy or coverage arms.

Evaluation. The encoder is frozen. Per-slice features are mean-pooled over the volume, followed by LayerNorm and a linear head. The head is selected by validation AUC and reported on the held-out test split. We report paired bootstrap confidence intervals over test volumes (10,000 resamples, stratified by class, the same resampled index set applied to every arm) and DeLong tests for correlated ROC curves [DeLong et al., 1988], both of which are valid here because all arms are evaluated on the same cases in the same order. Each test volume is a distinct subject, so case-level resampling is already subject-level and no cluster correction is required [Obuchowski, 1997, Rutter, 2000]. We follow Maier-Hein et al. [2024] in reporting the operating-point metrics alongside AUC rather than AUC alone (Appendix I).

Numerical precision. The probe harness defaults to mixed precision. The RANDOM, CENTROID and ENVELOPE long-horizon probes were fitted under that default (fp16); the ANATOMY-V1, ANATOMY-V2, COVER and ancestor probes were fitted with autocast explicitly disabled (fp32). Protocol is therefore not uniform across probes, so we partition the comparisons: all headline contrasts in Table 1 are drawn from arms sharing both epoch *and* precision, and any contrast that would cross the precision boundary is marked as such and excluded from headline claims. Section 5.5 measures the precision effect.

5 Results

Each policy was pretrained once, so the intervals below are sampling error over test subjects, not seed-to-seed variance (Section 6).

5.1 Guidance helps; anatomical precision does not reliably add

Table 1 gives the main comparison, Table 11 in Appendix L the full paired inference for all nine precision-matched contrasts, and Figure 2 the trajectories. All six contrasts against the null survive Benjamini–Hochberg correction over that family of nine ($q \leq 0.0299$ throughout). **H1 holds**: aiming the same rectangles at retinal tissue (ENVELOPE) beats unguided masking by $+0.0120$ at epoch 50 (CI $[+0.0068, +0.0173]$) and by $+0.0062$ at epoch 100 (CI $[+0.0010, +0.0113]$). Aiming is not containment: the rectangles are large, so only 43.5% of ENVELOPE's masked cells land on tissue (Table 2). Why the null nonetheless stays this close is examined in Appendix F.

H2 is not supported, but the anatomy result is mixed. H2 predicts the anatomy arms exceed ENVELOPE. Tested directly at the matched epoch 50, ANATOMY-V2 sits -0.0107 *below* ENVELOPE (CI $[-0.0167, -0.0046]$, $p = 0.0006$). At epoch 30, however, ANATOMY-V1 sits $+0.0044$ *above* it (CI $[+0.0009, +0.0078]$, $p = 0.0127$). These are different implementations at different epochs, so we do not read either as settling H2; what the pair rules out is a uniform advantage for anatomical precision, not the possibility that some shaped policy helps somewhere. Against the shared null, EN-

Table 1: Frozen MeanPool probe, FairVision test (N=3000, 1466 positive). All arms continue from one epoch-25 ancestor (AUC 0.8487). Δ is versus RANDOM at the *matched* epoch. The upper block is precision-matched (fp16 throughout) and carries every headline claim. The lower block is fp32, and each Δ is taken against an fp32 null re-probed at the same epoch, so it is matched on both epoch and precision. The measured fp16/fp32 gap is at most 2×10^{-4} (Appendix M).

		epoch 50		epoch 75		epoch 100	
policy	guidance signal	AUC	Δ	AUC	Δ	AUC	Δ
RANDOM	none (null)	0.8641	—	0.8723	—	0.8746	—
ENVELOPE	segmenter (location only)	0.8761	**+0.0120**	0.8803	**+0.0080**	0.8807	**+0.0062**
CENTROID	intensity centroid, no segmenter	0.8740	**+0.0099**	0.8836	**+0.0113**	**0.8855**	**+0.0109**
ANATOMY-V2	segmenter (target shape)	0.8654	**+0.0013**	0.8612	**−0.0111**	—	—
COVER f=.21	segmenter (coverage)	0.8643	**+0.0002**	0.8639	**−0.0084**	0.8577	**−0.0168**

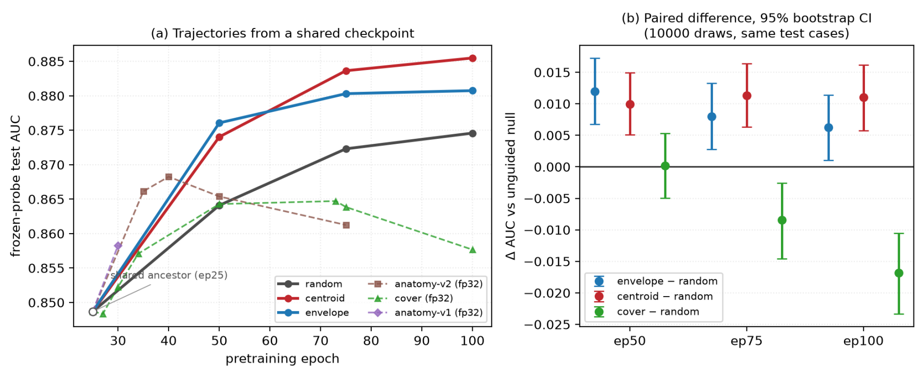

Figure 2: **(a)** Frozen-probe test AUC against pretraining epoch. All arms fork from the same epoch-25 checkpoint (open circle). Solid lines are the precision-matched fp16 family; dashed lines are fp32 arms and must not be compared vertically against the solid lines. **(b)** The actual inference: paired differences against the unguided null on identical test cases, with 10,000-resample bootstrap intervals. All six intervals exclude zero. Paired differences are plotted rather than per-arm error bars because marginal intervals carry between-case variance that cancels in a pairing and would understate the evidence.

VELOPE gains $+0.0120$ while ANATOMY-V2 gains $+0.0013$ and COVER $+0.0002$, neither separable from it; CENTROID gains $+0.0099$.

At the matched epoch 50 neither is *worse* than the null: ANATOMY-V2 (CI $[-0.0055, +0.0082]$, $p = 0.7153$) and COVER (CI $[-0.0050, +0.0053]$, $p = 0.9513$) are both indistinguishable from it. The data rule out monotone improvement with anatomical precision, not harm.

Carried to epoch 100 the coverage arm degrades. COVER is the only policy that uses the segmenter to choose *how much* to hide, and carried to the full horizon it produces the study's one *negative* result. It climbs normally to epoch 50, peaks at epoch 73 (0.8647), then *falls* to 0.8577, losing -0.0071 from its own peak (p<0.0001) while the unguided null keeps improving (Figure 2). The gap against the null widens with training: $+0.0002$ at epoch 50, -0.0084 at epoch 75, and -0.0168 at epoch 100 (CI $[-0.0233, -0.0105]$, p<0.0001).

We cannot read this as evidence that high coverage is harmful. An audit found that the collator shortens predictor targets *after* COVER chooses its placement, so its delivered masks hide 73.1% of anatomy — less than ENVELOPE's 77.6% (Figure 3) — and the arm never realised the coverage it was configured for; Appendix E gives the full account.

The strongest policy is the one that uses no segmentation model. CENTROID gives $+0.0109$ over the null at epoch 100 (95% CI $[+0.0057, +0.0160]$, p<0.0001, q =<0.0001), with a margin that is stable across training ($+0.0099$, $+0.0113$, $+0.0109$). It exceeds the segmenter-guided ENVELOPE arm at epoch 100 by $+0.0047$ (CI $[+0.0004, +0.0091]$, $p = 0.0294$), though the two are

Table 2: Measured mask geometry, 600 held-out slices, production samplers. "purity" is the fraction of masked cells lying on tissue; "loss slots" is the number of predictor targets contributing to the loss. AUC is quoted at the *matched* epoch 50, so the geometry-to-AUC association is not read across mixed epochs.

policy	anatomy hidden	purity	mask ratio	context kept	loss slots	AUC @ep50
RANDOM	52.2%	31.6%	43.7%	42.1%	157.7	0.8641
CENTROID	62.2%	41.1%	40.0%	45.6%	158.4	0.8740
ENVELOPE	76.9%	43.5%	46.4%	40.5%	159.9	**0.8761**
COVER	74.1%	45.3%	43.3%	43.5%	160.0	0.8643
ANATOMY-V2	80.3%	**97.3%**	21.4%	67.9%	64.0	0.8654

indistinguishable at epochs 50 and 75. ENVELOPE's margin over the null *decays* with training ($+0.0120 \rightarrow +0.0080 \rightarrow +0.0062$) while CENTROID's does not.

5.2 Mask geometry does not explain the ordering

To test H3 we ran each production sampler on 600 held-out slices and measured what it actually does (Table 2, Figures 13 and 14). Three observations follow.

First, **anatomical targeting does not explain the ordering**. Over the four rectangle arms the rank correlation between fraction-of-anatomy-hidden and AUC is positive (Spearman $+0.80$, $p = 0.20$), as is the correlation with mask *purity* at $+0.40$ ($p = 0.60$); including the anatomy arm these fall to $+0.50$ and $+0.20$. At n=4–5 arms none of them resolves the association in either direction, and we draw no inference from them. The direct comparison carries the point instead: the anatomy arm places 97.3% of masked cells on tissue and does not separate from the null; CENTROID places 41.1% on tissue and reaches the highest epoch-100 AUC. Occlusion attribution on the fine-tuned probes is correspondingly diffuse (Appendix H).

Second, **the anatomy arms are confounded on several axes simultaneously**. They hide only 21.4% of the grid against 40–46% for the rectangle arms, leave 67.9% of tokens visible as context against 40–46%, and supply 64 predictor loss slots against 158–160. Any of these could account for their deficit without invoking target shape at all. **H3 is therefore not identified by this design**, and we do *not* claim that irregular target shape is harmful. Figure 15 makes the four-way divergence explicit.

These are not consequences of blob geometry: the anatomy arms resample every target to a fixed K=16 cells, so $4 \times 16 = 64$ loss slots follows by construction, while the rectangle arms do not set K. The two families are collated differently, which weakens "everything else held fixed" for the anatomy-versus-rectangle comparison specifically; contrasts within the rectangle family are unaffected (Appendix E).

Third, CENTROID is the most *spatially consistent* policy we measured. Its per-cell hit map is the most concentrated of any arm (Gini 0.400 versus 0.316–0.338 for the other rectangle arms): it repeatedly asks the encoder to represent the same region of the image. This suggests a mechanism other than anatomical correctness — consistency of the predictive task — which we return to in Section 6.

5.3 The advantage concentrates where labels are scarce

A masking policy that only helps when labels are plentiful is of limited clinical use, so we withdrew supervision and re-fitted the same probe. Protocol and label subsets are held fixed — the subsampling seed depends on the fraction and repeat but never on the arm — and at full supervision this reproduces Table 1 to within 0.0003.

The observed margin is larger with fewer labels. At 5% of the labelled set (n=300) CENTROID leads the null by $+0.0496$ (0.8335 against 0.7839), against $+0.0108$ at full supervision — more than four times the advantage. At 1% the separation is wider again (0.7576 against 0.6961), though the spread across repeats (0.0359–0.0368) is large enough that we read the ordering and not its size. Whether the gap genuinely widens as labels are withdrawn we did not test; the ordering across fractions is a point-estimate observation. The absolute gains at full supervision are small. Appendix G gives the full curve.

5.4 Subgroups: the ordering never changes

We audit seven stratifications across 23 probes spanning aggregate AUC 0.8483 to 0.8855. Predictions are joined to the released FairVision metadata in deterministic loader order, validated by exact label reconstruction (3000/3000 cases, every probe). Probes that the exclusion rules of Appendix D remove are removed here too.

The ordering of subgroups never changes. The worst-performing group is the same in every one of the 23 probes for sex (female), race (black), ethnicity (hispanic) and disease severity (mild (-6,-2]). No masking policy reorders them. The probes share a test set, an epoch-25 ancestor and a probe seed, and several are different epochs of one branch, so they are *not* independent; we report this as a descriptive regularity and attach no p-value. What it shows is that the disparity is not introduced by any policy we tried.

Point estimates rise everywhere; few gaps reliably move. A widening max–min gap is easy to misread as harm. It is not harm. At epoch 100 the strongest policy raises the point estimate for *every* race stratum over the null, and the max–min gap widens only because the already-best asian subgroup gains most. Paired resampling, however, separates only the white subgroup's gain from zero ($+0.01129$, CI [$+0.00518$, $+0.01731$]); the black ($+0.01467$, CI [-0.00055, $+0.03060$]) and asian intervals include it, the smaller strata being noisier. The gains are therefore consistent in sign across race, not provably positive in every group. Nor is there a defensible gap *trend*. Across checkpoints the race gap correlates with aggregate AUC at $\rho = +0.473$ ($p = 0.0225$, $q = 0.0668$ after Benjamini–Hochberg over the seven attributes tested), but those checkpoints are not independent: they are 23 probes drawn from only 7 pretraining branches, so a branch probed at four epochs contributes four correlated points. Collapsing to one point per branch, the association disappears ($\rho = +0.321$, $p = 0.4821$). The checkpoint-level correlation is pseudo-replicated and does not show that better models are less fair. The same holds for severity in the opposite direction ($\rho = -0.449$ per checkpoint, -0.821 per branch). Only the sex gap is consistent across both views, and it *narrows* ($q = 0.0038$; branch-level $\rho = -0.821$, $p = 0.0234$). Neither harm nor benefit to equity is established here.

Every severity stratum improves; differential benefit is unresolved. Scoring each severity stratum against the shared pool of all negatives, the strongest policy improves mild disease by $+0.0137$ at matched epoch 100, moderate by $+0.0102$, severe by $+0.0063$. These are paired differences on the same subjects and all three intervals exclude zero (mild [$+0.00541$, $+0.02192$], severe [$+0.00104$, $+0.01187$]), so each stratum improves. Whether mild improves *more* we did not test: that needs a paired contrast between the gains, not three contrasts against zero. What does not move is the difficulty ordering itself: the severe-to-mild gap stays within 0.0114 of constant across all 23 probes while aggregate AUC spans 0.8483–0.8855. For equity the quantity that matters is the paired *difference* between groups' gains; between the worst- and best-served race strata it is -0.00091 (CI [-0.02379, $+0.02139$]) and by sex $+0.00146$ (CI [-0.00853, $+0.01156$]). Both contain zero, so at these subgroup sizes we cannot resolve whether the improvement is evenly distributed. Appendix C gives every stratum. No policy here is a fairness intervention and none should be presented as one.

5.5 Controls

A full re-fit of the probe pipeline at fp32 shifts every arm by less than 2×10^{-4}, orders of magnitude below the effects in Table 1, so the differing autocast settings of Section 4 are not the effect and every contrast is matched on precision (Appendix M). Every number quoted here is generated by one script from stored per-case predictions; that script, the epoch-100 CENTROID encoder, its head and those predictions are released, links withheld for anonymity. Re-encoding the test split from those weights on different hardware, at fp32 rather than fp16 and with a different encoder chunk size, reproduces the reported AUC to 9.8×10^{-6} (0.8854754 versus 0.8854852): 22 discordant pairs out of 2,248,844.

6 Discussion and limitations

What the evidence supports. Guidance helps: restricting targets to tissue is worth $+0.0120$ at the matched epoch, and a consistent intensity-located band is worth $+0.0109$ at epoch 100, with paired intervals excluding zero at every epoch measured. Using the segmenter more aggressively — to shape targets or to maximise anatomy coverage — does not add to that and can subtract. For *this* task the segmentation stage bought no more than location, and a one-line intensity statistic recovered the whole benefit. We would not generalise that to tasks where anatomy is the output, such as layer segmentation or lesion localisation.

What it does not support. We cannot claim anatomy-shaped masking is *harmful*: at the matched epoch it sits $+0.0013$ from the null with an interval ($[-0.0055, +0.0082]$) that comfortably contains zero, so it is indistinguishable rather than worse. Nor is target *shape* identified as the operative variable (Table 2), nor is CENTROID's mechanism. Two explanations remain live: consistency of location (the encoder is repeatedly forced to represent the same region), and masking ratio or task difficulty. Distinguishing them requires an arm that randomises the band's vertical position while holding area and context fixed, which we did not run.

One pretraining run per policy. The paired bootstrap quantifies sampling error on a fixed test set; it does not quantify seed-to-seed variance of pretraining. With $n=1$ continuation per arm the *ranking* is not statistically established even where individual pairwise differences are significant on this test set. Bouthillier et al. [2021] show that single-seed comparisons routinely mistake optimisation noise for method effects. We can bound one component: with five probe seeds on two fixed encoders the probe-seed standard deviation is 0.0003 and 0.0018, and the arms stay fully separated across all five, so probe noise is three to seven times below the effects in Table 1. Those are *technical* replicates and do not estimate pretraining noise, which would need at least three paired continuations from the shared checkpoint with the continuation as the unit of replication. That is the single most valuable follow-up.

Scope, coverage and an incomplete arm. All results are glaucoma classification from FairVision OCT; generality to other modalities, diseases, or segmentation-quality regimes is untested, and the anatomy arms depend on MIRAGE's prediction quality on this data, so a better segmenter might change the conclusion. Only $f=0.21$ was pretrained for COVER, so we report no dose-response over the visible-tissue floor and make no claim that any floor is optimal. That arm reaches epoch 100, but the collation defect above (Appendix E) means its trajectory is not evidence about aggressive coverage. Its epoch-50 value is retained as a matched-epoch comparison against every other arm. Finally, one dataset and one test split were reused throughout this programme's history, and policies, checkpoints and analyses were chosen after repeated inspection of that split; the intervals we report are therefore best read as descriptive rather than as confirmatory inference.

Broader impact and ethics. This study uses a public, de-identified retinal dataset under its release terms and involves no new data collection or human subjects. No model reported here is clinically validated or intended for deployment. Appendix J gives the full statement.

7 Conclusion

We tested the intuition that masked predictive pretraining in medical imaging should target clinically meaningful anatomy. In these runs, under a matched schedule, effective batch size, and shared ancestor checkpoint, guidance helped — but anatomical precision did not. Restricting targets to retinal tissue improved AUC over unguided masking by $+0.0120$ at the matched epoch, while shaping targets to trace the segmented retina matched unguided masking at epoch 50 and fell below it by epoch 75, and the best policy consulted no segmentation model. A subgroup audit shows every disease stage improving with intervals excluding zero, race gains consistent in sign, and the difficulty ordering untouched. We suggest the operative variable is the consistency of the predictive task rather than its anatomical correctness.

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A All frozen probes

Table 3 lists every frozen MeanPool probe in the study.

Table 3: Every frozen probe in the study, including runs excluded from the analysis and the reasons. The probe protocol is otherwise the same for every row — FairVision test N=3000, seed 42, frozen encoder, MeanPool + LayerNorm + Linear — and differs only in the precision column. "Precision" is the probe-time numerical precision, inferred from the stored prediction dtype, and is distinct from the pretraining target precision discussed in Appendix D.

policy	epoch	precision	test AUC	95% CI
ANATOMY-V1	30	fp32	0.8583	[0.8450, 0.8712]
ANATOMY-V2	35	fp32	0.8661	[0.8531, 0.8787]
ANATOMY-V2	40	fp32	0.8683	[0.8553, 0.8807]
ANATOMY-V2	50	fp32	0.8654	[0.8522, 0.8781]
ANATOMY-V2	75	fp32	0.8612	[0.8477, 0.8740]
ancestor	25	fp32	0.8487	[0.8349, 0.8621]
COVER	27	fp32	0.8483	[0.8346, 0.8617]
COVER	30	fp32	0.8522	[0.8386, 0.8654]
COVER	34	fp32	0.8571	[0.8437, 0.8700]
COVER	50	fp32	0.8643	[0.8513, 0.8768]
COVER	73	fp32	0.8647	[0.8517, 0.8772]
COVER	75	fp32	0.8639	[0.8507, 0.8763]
COVER	100	fp32	0.8577	[0.8443, 0.8707]
ENVELOPE	30	fp32	0.8539	[0.8405, 0.8669]
ENVELOPE	50	fp16	0.8761	[0.8635, 0.8879]
ENVELOPE	50	fp32	0.8761	[0.8635, 0.8879]
ENVELOPE	75	fp16	0.8803	[0.8677, 0.8922]
ENVELOPE	75	fp32	0.8803	[0.8677, 0.8922]
ENVELOPE	100	fp16	0.8807	[0.8683, 0.8926]
ENVELOPE	100	fp32	0.8808	[0.8683, 0.8927]
CENTROID	50	fp16	0.8740	[0.8615, 0.8862]
CENTROID	50	fp32	0.8740	[0.8614, 0.8862]
CENTROID	75	fp16	0.8836	[0.8714, 0.8955]
CENTROID	100	fp16	0.8855	[0.8735, 0.8971]
CENTROID	100	fp32	0.8853	[0.8732, 0.8971]
RANDOM	50	fp16	0.8641	[0.8510, 0.8769]
RANDOM	50	fp32	0.8641	[0.8511, 0.8769]
RANDOM	75	fp16	0.8723	[0.8593, 0.8849]
RANDOM	75	fp32	0.8723	[0.8593, 0.8849]
RANDOM	100	fp16	0.8746	[0.8618, 0.8869]
RANDOM	100	fp32	0.8745	[0.8617, 0.8868]
anatomy-v2	75	fp32	0.8625	*excluded*
anatomy-v2	92	fp32	0.8604	*excluded*
random	30	fp32	0.8558	*retracted*
random	50	fp32	0.8590	*retracted*
random	75	fp32	0.8612	*retracted*
random	100	fp32	0.8607	*retracted*

B Masking policies across the volume

Figure 1 shows one B-scan for legibility. Figure 4 repeats the same rendering on five B-scans sampled across the depth of a volume, so that the stability of each policy under changing retinal geometry can be inspected. The samplers, the colour assignment, and the anatomy contour are identical to Figure 1.

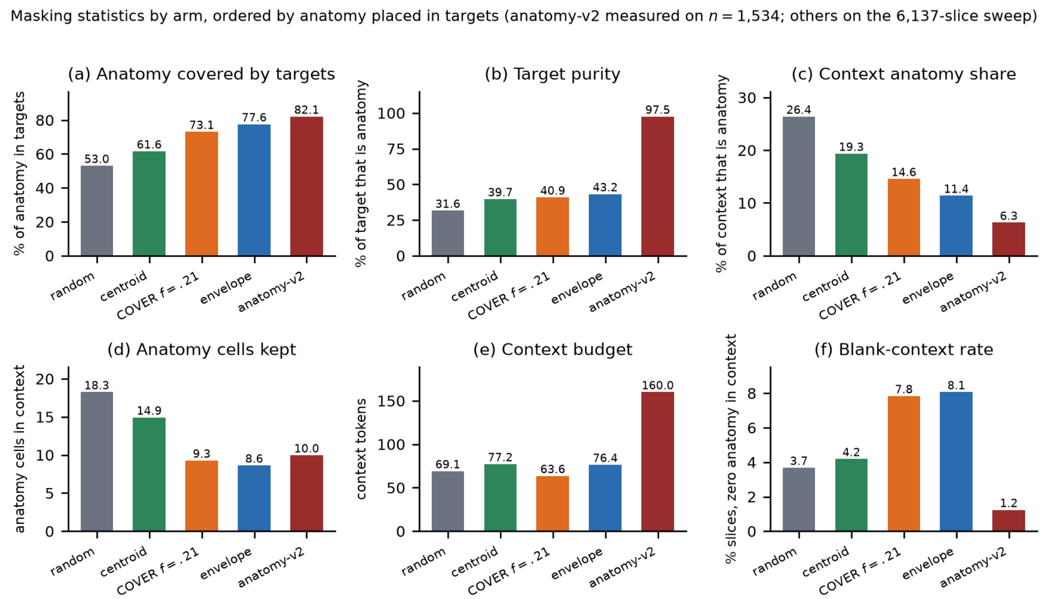

Figure 3: What each sampler actually does, measured on the 6,137-slice sweep (ANATOMY-V2 on n=1,534). Panel (a) is the quantity the COVER arm was built to maximise: it reaches 73.1%, *below* ENVELOPE's 77.6%, which is the discrepancy Appendix E explains. Panel (b) shows the ordering this paper tests: target purity rises monotonically from RANDOM to ANATOMY-V2, while downstream AUC does not follow it. Panels (c)–(f) give the confounds we do not disentangle — how much anatomy survives in the context, the context budget, and how often a slice is left with no anatomy visible at all.

C Subgroup and severity analysis

All numbers below come from saved test predictions joined to FairVision's released metadata in deterministic loader order; the join is validated by exact label reconstruction (3000/3000 labels recovered in every probe). The 23 probes used here are exactly those the inventory marks valid: the four retracted coverage probes of Appendix D and the two precision-spliced ANATOMY-V2 probes are excluded here as everywhere else in the paper.

Why we report two sample sizes. The 23 probes span only 7 pretraining branches, because several are different epochs of one training run. Probes within a branch share an encoder lineage, a seed and the test split, so treating them as 23 independent observations is pseudo-replication. Table 4 therefore reports each trend twice: once per checkpoint, and once after averaging within each branch. Where the two disagree, the branch-level figure is the one to believe.

Race. The black subgroup has the lowest AUC in every one of the 23 probes (n=431 black, n=251 asian, n=2318 white; Figure 5). At matched epoch 100 the black subgroup improves from 0.8325 under RANDOM to 0.8472 under CENTROID (+0.0147), a larger absolute gain than the white subgroup's (0.8741 $\rightarrow$ 0.8853). The max–min gap nonetheless widens slightly, because the asian subgroup — already the highest — gains most (0.9033 $\rightarrow$ 0.9189). A widening max–min gap with universally rising subgroup AUCs is not evidence of harm. The checkpoint-level correlation between aggregate AUC and racial gap is ρ=+0.473 (p=0.0225). It passes Benjamini–Hochberg (q=0.0668) but vanishes once probes are aggregated to their pretraining branches ($\rho = +0.321$, p=0.4821), and the checkpoint-level test is pseudo-replicated because the 23 probes come from only 7 branches.

Disease severity. FairVision's label is defined by visual-field mean deviation (md): every test volume with md ≤ -2 is positive. Severity therefore cannot be used as an ordinary subgroup axis, because within-bin AUC is undefined — a trap that silently yields nothing under naive stratification. Instead each positive stratum is scored against the shared pool of all 1534 negatives (Table 5).

Table 4: Subgroup gap trends. “gap range” is the max–min subgroup AUC gap observed across the 23 valid probes. q is Benjamini–Hochberg across the seven attributes tested, which is necessary because seven trends were examined. Three attributes survive correction at checkpoint level: sex and disease severity, both of which *narrow* as aggregate AUC rises, and race, which widens. Race is the one attribute whose checkpoint-level trend does not persist under branch aggregation (p=0.4821), where the effective sample is 7 independent branches rather than 23 probes; we therefore draw no trend conclusion for race.

			per checkpoint (n=23)			per branch (n=7)	
attribute	worst group	gap range	ρ	p	q	ρ	p
sex	female (23/23)	0.0272–0.0407	-0.664	0.0005	0.0038	-0.821	0.0234
race	black (23/23)	0.0391–0.0935	+0.473	0.0225	0.0668	+0.321	0.4821
ethnicity	hispanic (23/23)	0.0180–0.0661	+0.435	0.0381	0.0668	+0.679	0.0938
language	other languages (19/23)	0.0518–0.0805	+0.311	0.1483	0.1730	+0.786	0.0362
marital status	single (23/23)	0.0286–0.0654	-0.210	0.3351	0.3351	-0.679	0.0938
age	$<$60 (22/23)	0.0430–0.0704	+0.399	0.0591	0.0828	+0.357	0.4316
disease severity	mild $(-6, -2]$ (23/23)	0.1286–0.1400	-0.449	0.0318	0.0668	-0.821	0.0234

Table 5: Severity-stratified AUC at matched epoch 100. Each positive stratum is scored against the shared pool of all 1534 negatives. Every stratum improves; the gains were not tested against one another.

stratum	n_+	RANDOM	ENVELOPE	CENTROID	Δ
severe (md ≤ -12)	334	0.9525	0.9559	0.9588	+0.0063
moderate ($-12 <$ md ≤ -6)	460	0.9030	0.9103	0.9131	+0.0102
mild ($-6 <$ md ≤ -2)	672	0.8164	0.8232	0.8301	**+0.0137**

The severe-to-mild gap is 0.1361 for the null and 0.1286 for the strongest policy, and stays within 0.0114 (0.1286–0.1400) across all 23 probes while aggregate AUC spans 0.8483–0.8855. Target placement raises the whole severity curve, all three strata separated from zero under paired resampling (Table 6), the largest point estimate at mild disease ($+0.01372$, CI $[+0.00541, +0.02192]$) and the smallest at severe ($+0.00626$). That ordering is descriptive only: we did not test the paired *difference* between strata, so it does not establish that mild disease benefits most. We note that mild functional loss is harder to separate from normal OCT by construction, so the difficulty ordering is expected clinically; what the data add is that no masking policy we tested changes it.

Table 6: Paired change in subgroup AUC from RANDOM to CENTROID at matched epoch 100. Each bootstrap draw resamples subjects once and applies the same resampled set to both arms and every stratum, so these are paired differences rather than differences of independent estimates.

stratum	n	Δ AUC	95% CI	excludes 0
severity, mild	2206	+0.01372	[+0.00541, +0.02192]	yes
severity, moderate	1994	+0.01016	[+0.00262, +0.01757]	yes
severity, severe	1868	+0.00626	[+0.00104, +0.01187]	yes
race, White	2318	+0.01129	[+0.00518, +0.01731]	yes
race, Black	431	+0.01467	[-0.00055, +0.03060]	no
race, Asian	251	+0.01558	[-0.00079, +0.03283]	no
sex, Female	1716	+0.01152	[+0.00412, +0.01890]	yes
sex, Male	1284	+0.01006	[+0.00335, +0.01678]	yes

Limitations. The p-values here are descriptive, for the reasons given in Section 6. Race and ethnicity are self-reported categories inherited from FairVision’s coding; at these subgroup sizes we cannot speak to intersectional groups. All figures are frozen mean-pool linear probes. Subgroup sensitivity at a transferred threshold is reported in Appendix I, but we report no subgroup calibration and no intersectional breakdown, so this is not a complete fairness evaluation. Disparities could differ under fine-tuning or a stronger head.

D Excluded runs

Four probes of an earlier *null* run are excluded from all analysis. They are unguided-masking controls from a superseded coverage campaign — the inventory lists them under RANDOM, and their run tags begin `frozen_cover_random` because of that campaign, not because they use a coverage policy. They were trained with half-precision EMA targets and a different encoder-truncation policy from the arms they would be compared against, so their deficit is not attributable to masking. Similarly, the ANATOMY-V2 probes at epochs 75 and 92 follow a change of target precision partway through that arm's training; they are listed in Table 3 but excluded from the matched-epoch comparisons in Table 1.

E A collation defect in the coverage arm, and what it invalidates

We audited the COVER sampler after its epoch-100 result and found a defect. It invalidates the reading of that arm and weakens one cross-family comparison.

The defect. Predictor targets are collated by truncating every target to the length of the shortest target anywhere in the microbatch. COVER chooses its placement *before* this happens: it greedily maximises hidden anatomy subject to a visible-tissue floor, evaluated on full rectangles, and the collator then shortens those rectangles. The optimisation is therefore defeated after the fact.

Measured consequences. On held-out slices, COVER's placement hides 78.6% of anatomy mass, close to its configured target, but the masks actually delivered hide 73.9%; an independent sweep over $6{,}137$ slices gives 73.1% against 77.6% for ENVELOPE. The arm intended to hide *more* anatomy than ENVELOPE in fact hides *less*. Only 32.5% of images retain all four rectangular targets, the mean being 2.51 of 4, and across $24{,}000$ emitted targets only 73.4% remain perfect rectangles. Because target cells are enumerated row-major, truncation keeps the top of each rectangle and discards the bottom, so the loss is directional rather than a neutral shrink.

Why it went unnoticed. Realised coverage is recorded before truncation. Every training log therefore reported the intended figure, and the instrument agreed with the design rather than with the data.

What it invalidates. We cannot claim that COVER tests aggressive anatomy coverage, nor that its decline shows over-covering anatomy is harmful; on delivered masks it does not out-cover ENVELOPE. Its measured trajectory remains a real property of the policy as implemented. The mask geometry in Table 2 is unaffected, having been measured on delivered masks.

Scope. ENVELOPE shares the same collation path and the same directional clipping, but sets no exact coverage target, so it has no comparable optimisation to defeat. The anatomy arms instead resample every target to a fixed $K{=}16$ cells. Contrasts within the rectangle family are collated identically and are unaffected; the anatomy-versus-rectangle contrast is not, and we treat it accordingly in Section 5. A corrected coverage arm must be retrained from the shared ancestor, since patching mid-trajectory would splice two masking policies into one curve.

Status of a corrected run. At submission the corrected arm is not run. Until it exists, the question this arm was built to answer — whether hiding most of the anatomy is harmful — remains *open*.

F Why unguided masking is such a strong baseline

Unguided masking spends most of its predictor targets on background, yet it reaches within $+0.0109$ of the best policy. We investigated whether background carries diagnostic signal, and whether position embeddings are the mechanism.

Background is a real *pretraining* signal. At epoch 100 the random arm's predictor beats a per-position, no-context reference by 0.680 on background targets, above its 0.633 on anatomy targets: predicting a background token requires more than knowing where it is. Pretraining also spends capacity reorganising background — background self-similarity falls from 0.784 untrained to 0.346 at epoch 100.

But it does not help the classifier. Pooled background features are 95.2% linearly reconstructible from pooled anatomy features. Residualised, background predicts glaucoma at test AUC 0.5515 (CI $[0.5165, 0.5893]$). Appending background to anatomy *lowers* test AUC by 0.0076 (CI $[-0.0139, -0.0012]$) for the random arm and moves it under ± 0.002 for every other. A background-only probe scores 0.867; self-attention mixes tissue information into background tokens, so that figure cannot be read as background signal, and we did not measure how much mixing occurs.

Position dominates background tokens, but is not the whole story. At layer 0, 90.8% of the across-position variance of the encoder input at background cells comes from `pos_embed`, against 40.8% at tissue cells (stable across six thresholds and both checkpoints). Mechanically a background token is close to its position code. That the predictor still beats a per-position reference shows the pretraining signal is not purely positional. The causal chain from this mechanism to downstream AUC remains open.

G Label efficiency

We measured each policy as supervision is withdrawn (Section 5.3). We subsample the labelled training set to fractions of its size, fit the same frozen linear probe on cached epoch-100 features, and repeat 5 times per fraction. The per-fraction seed depends only on fraction and repeat, never on arm, so all arms see identical label subsets and the comparison is paired.

Protocol parity. This sweep uses the *same* probe protocol as Table 1 — minibatches of 256, AdamW at 4×10^{-4}, weight decay 0.05, 50 epochs with 5 warmup and cosine decay, and the epoch selected on validation AUC. An earlier version of this experiment used a cheaper full-batch fit, which put the null at 0.8811 at full supervision against 0.8746 in the primary protocol and so made the two tables disagree. Under matched protocols they now agree to within 0.0003 at full supervision (0.8748 against 0.8746 for the null, 0.8856 against 0.8855 for CENTROID), so the curve below and the headline table describe one probe, not two.

The advantage of guided masking is concentrated where labels are scarce (Table 7, Figure 6). At 5% of labels ($n{=}300$) CENTROID leads the null by $+0.0496$ (0.8335 against 0.7839), against $+0.0108$ at full supervision — more than four times the margin. At 1% the gap is wider still (0.7576 against 0.6961), though the standard deviations there (0.0359–0.0368 across arms) are large enough that we read the ordering but not its precise size.

The defective COVER arm (Appendix E) is the worst arm at every fraction, reaching only 0.6879 at 1% against 0.6961 for the null, consistent with its targets being degraded rather than merely differently placed.

Table 7: Label efficiency: frozen-probe test AUC against the fraction of the labelled training set used. All values measured, 5 repeats per fraction except at full data, which is deterministic.

labels	n	RANDOM	CENTROID	ENVELOPE	COVER
1%	60	0.6961	0.7576	0.7471	0.6879
5%	300	0.7839	0.8335	0.8197	0.7484
10%	600	0.8169	0.8490	0.8404	0.7743
25%	1500	0.8497	0.8732	0.8654	0.8162
100%	6000	0.8748	0.8856	0.8813	0.8586

H Occlusion attribution: where the classifier looks

The analyses in this appendix were run on the three *fine-tuned* probes (mean-pool, cross-attention pool, and a depth-1 attentive probe) that tie at test AUC ≈ 0.887, not on the frozen probes used for the arm comparison in the main text. They characterise what the downstream classifier uses, and they are reported here because two of the findings are cautionary rather than positive.

Method. Attribution is by *occlusion*, which is architecture-agnostic and interventional on the model. It measures the model's sensitivity to deleting a token, not a causal effect of anatomy on disease. Gradient$\times$input is not used because two of the three probes are non-linear in the slice-token matrix, where gradient attribution is misleading. For slice-level attribution each slice token is zeroed in turn and Δlogit recorded (Figure 7); for window-level, seven consecutive slices (Figure 8); for patch-level, leave-one-out over the 256 patch tokens of a target slice.

A bimodal attribution curve that is an artefact, not anatomy. The population-averaged curve peaks at native slice ≈ 63 and ≈ 137 with a dip near 95. The tempting reading — that the model has discovered the superior and inferior optic disc rim — is **wrong**, and we report it because it is exactly the kind of claim a plausible-looking attribution figure invites. Three tests reject it. First, if the two peaks were bilateral anatomy then within a diseased eye they should co-occur, giving positive per-volume correlation between them; the observed correlation is slightly *negative* (-0.22, -0.07, -0.14 across the three probes). Second, clustering the per-volume curves into two groups yields clusters that are near-perfect *mirror images* along the slice axis ($\mathrm{corr}(c_1, \mathrm{flip}(c_2)) = 0.971$ and 0.988 for the two well-structured probes, against raw correlations of -0.124 and -0.478). Third, using that cluster label as pseudo-laterality and flipping one cluster collapses the bimodality asymmetrically (Figure 9). The signature is OD/OS axial storage: each eye carries its dominant signal on one side of the disc, right and left eyes are stored with opposite axial orientation, and population averaging manufactures an illusion of symmetry.

Errors are weaker signal, not different anatomy. Stratifying by outcome, false-negative curves are scaled-down true-positive curves with the same shape, and false positives likewise mirror true negatives (Figure 10). Attribution structure is also near-invariant to prediction confidence ($|r| \leq 0.25$ between peak contribution magnitude and $|\mathrm{logit}|$). The error rate at threshold 0.5 therefore reflects signal-strength saturation on hard cases, not the model attending to the wrong structures.

Diffuse attribution supports the main claim. The classifier's evidence is spread across the middle two-thirds of the volume and across most patches within a slice (Figure 11), rather than concentrated on a compact anatomical target. A masking policy that sharpens targets onto a precise anatomical boundary is therefore optimising for a spatial prior the downstream task does not appear to need, which is consistent with Section 5.2 finding no relationship between anatomical specificity and downstream AUC. The OD/OS result is a methodological caution: a figure that looks anatomically meaningful can be an artefact of data storage.

I Clinical operating points and calibration

AUC is threshold-free, but a screening tool is deployed at a threshold. The threshold is selected on the RANDOM arm's *validation* split at a fixed target specificity and then applied unchanged to every arm on the test split: one shared deployed threshold rather than a per-arm one, which simulates a fielded screen and is conservative for the guided arms, since they are not permitted to retune. The achieved test specificity shows whether the threshold survives the shift. Cohort prevalence is 0.4887.

Table 8: Operating points at matched epoch 100. One threshold, chosen on the RANDOM validation split to hit the target specificity, is applied unchanged to all arms on test. ECE is a 15-bin expected calibration error.

policy	target spec.	sens.	spec. (test)	PPV	NPV	Brier	ECE
RANDOM	0.85	0.7701	0.8259	0.8087	0.7899	0.1416	0.0355
RANDOM	0.90	0.7162	0.8794	0.8502	0.7643	0.1416	0.0355
ENVELOPE	0.85	0.7735	0.8351	0.8176	0.7942	0.1364	0.0361
ENVELOPE	0.90	0.7306	0.8807	0.8541	0.7738	0.1364	0.0361
CENTROID	0.85	0.7851	0.8318	0.8169	0.8020	0.1341	0.0320
CENTROID	0.90	0.7428	0.8696	0.8448	0.7797	0.1341	0.0320

Two observations (Table 8). First, the threshold transfers imperfectly: a validation target of 0.90 yields test specificity 0.8696 to 0.8807 across the three arms, so the operating point moves by roughly one point between arms and the comparison below is at a shared threshold rather than at a shared false-positive rate. Second, the AUC advantage survives as a usable sensitivity gain. At a target specificity of 0.90, CENTROID detects 0.7428 of cases against 0.7162 for the unguided null, a paired difference of +0.0266 (CI [+0.0136, +0.0396], excluding zero), with the specificity caveat above. At a target of 0.85 the gain is +0.0150 (CI [+0.0020, +0.0280]). CENTROID is also the best-calibrated arm (Brier 0.1341 versus 0.1416; ECE 0.0320 versus 0.0355), which matters because a screening score that is used as a probability must be trustworthy as one.

A +0.011 AUC difference is hard to interpret clinically, and the ROC curves are nearly superimposed (Figure 12), whereas a change in cases detected at a fixed threshold is not. We state that comparison carefully: transferring one threshold moves achieved specificity from 0.8794 to 0.8696, so the arms are *not* compared at an identical false-positive rate and the sensitivity change partly reflects that shift. A deployment-grade comparison would select each arm's threshold on validation to meet the same specificity, lock it, and evaluate on an untouched cohort. One dataset, one split, one pretraining run per policy.

Who receives the benefit at the deployed threshold. Subgroup AUC is also threshold-free, so it cannot answer the question a deployment actually raises: a screen ships *one* threshold shared by everyone, and a model can lift every subgroup's AUC while delivering the gain unevenly at that single operating point. We therefore fix the threshold on validation at specificity 0.90 (0.5552), transfer it unchanged, and measure the change in sensitivity within each stratum (Table 9). Overall sensitivity rises from 0.7162 to 0.7428.

The point estimates alone suggest the gain is concentrated in the largest group: +0.0343 for white patients against +0.0037 for black patients, an apparent ninefold difference. **The intervals do not support that reading.** The black interval ([−0.0261, +0.0336]) reaches almost exactly the white point estimate, so the two are not separated; what distinguishes them is positive count (1080 versus 268), not demonstrated benefit. Only the white, female and male gains exclude zero, and those are the three largest strata. This is a *power* limitation of the cohort: the study cannot certify that the smaller strata benefit, and equally cannot show they do not. Certifying subgroup benefit at a deployed threshold would need a cohort with more positives in the smaller strata.

Table 9: Change in sensitivity from RANDOM to CENTROID at one shared deployed threshold (validation-selected, specificity 0.90), by stratum. Paired bootstrap over positives within each stratum. Epoch 100.

stratum	positives	Δ sensitivity	95% CI	excludes 0
Asian	118	+0.0085	[−0.0339, +0.0508]	no
Black	268	+0.0037	[−0.0261, +0.0336]	no
White	1080	+0.0343	[+0.0194, +0.0491]	yes
Female	811	+0.0259	[+0.0074, +0.0444]	yes
Male	655	+0.0275	[+0.0092, +0.0458]	yes

J Broader impact and ethics

This study uses a public, de-identified retinal dataset (FairVision) under its release terms and involves no new data collection and no human subjects. No model reported here is clinically validated or intended for deployment: the frozen-probe AUCs are research measurements on one dataset and one task, and are not evidence of clinical utility.

The subgroup analysis in Section 5.4 is a caution, not a contribution. Every masking policy we tested leaves the same groups worst-served, and the best-performing policy narrows none of them reliably, even though every subgroup point estimate rises. We would regard deployment of any encoder reported here, without a dedicated and prospective fairness evaluation on the target population, as inappropriate. We also stress that the audit is incomplete: beyond AUC and the transferred-threshold sensitivity of Appendix I, it contains no subgroup calibration, no predictive-value analysis and no intersectional breakdown.

Releasing pretrained weights carries the usual dual-use consideration that they may be applied to populations unlike the training cohort. We document the cohort composition (Appendix C) so that such a mismatch is detectable by anyone reusing the weights. The demographic attributes are self-reported categories inherited from the dataset's coding; they are used only for the subgroup audit and are never model inputs.

K Published ablations on informed versus random masking

Our result should be read against the following published comparisons (Table 10), which we collected while positioning this work. The pattern is that random masking is a strong baseline and informed masking wins only under some objectives and protocols; the deficit is largest under frozen evaluation.

Table 10: Published ablations comparing a random or simple baseline against an informed or structured masking scheme. Numbers are as reported by the cited papers; metrics differ across rows and are not comparable between rows.

study	baseline	informed variant	interpretation
MAE [He et al., 2022]	random-75: 84.9 FT / 73.5 lin	block-75: 82.8 / 63.9; grid-75: 84.0 / 66.0	random decisively better
SimMIM [Xie et al., 2022]	83.0 top-1	best block 82.7; square 82.6	random modestly better
SemMAE [Li et al., 2022]	random 66.8 lin	within-part 66.5; whole-part 52.9; adaptive 68.7	naive semantics hurt; scheduled semantics help
SemMAE part quality [Li et al., 2022]	no parts 63.7	raw parts 63.6; refined 65.0	an imperfect semantic model adds nothing
AutoMAE [Chen et al., 2023a]	MAE 83.26 FT	AutoMAE 83.32	effectively tied
HPM [Wang et al., 2023b]	random 82.49	moderate 82.95; hardest-only 81.40	benefit only with retained randomness
AttMask [Kakogeorgiou et al., 2022]	random 43.4 k-NN	high 43.5; hint 43.6	nearly indistinguishable
AttG-MAE [Sick et al., 2025]	MAE 78.5 @10% labels	guided 78.1	guidance loses in this protocol
SSiT [Huang et al., 2024]	no saliency 79.48 κ	25% mask 77.53; 50% 79.97	dataset-dependent, non-monotonic

Read together, these say that the *direction* of our finding is not new. What we add is the controlled measurement, not the sign of the result.

L Full paired-contrast table

M Full fp32 re-probe

The RANDOM, CENTROID and ENVELOPE long-horizon probes were originally fitted under the harness default, which enables autocast. To remove any doubt that the headline effect is a precision artifact, we re-ran the complete probe pipeline for those arms with autocast disabled, so that feature extraction, head optimisation and test evaluation are all fp32, under the exact protocol already used by the ANATOMY and COVER arms (Table 12). The encoder checkpoint is SHA-256 hashed before and after each run and verified unchanged, so no result here can be contaminated by an accidental encoder update.

N Fine-tuning narrows but does not erase the gap

Under full fine-tuning rather than a frozen probe, CENTROID reaches 0.8947 (mean-pool head) against RANDOM's 0.8868, a gap of $+0.0079$ compared with $+0.0109$ frozen. Taking each arm's best head instead, the gap is $+0.0069$. The pretraining difference is partially, but not entirely, absorbed by supervised adaptation — consistent with the masking policy shaping the representation rather than merely the optimisation start point. These runs used a different head and schedule from the frozen probes and are never pooled with them.

Table 11: Primary contrasts: matched epoch *and* matched precision. Δ is a paired difference on identical test cases; the interval is a 10,000-resample paired bootstrap and p is a DeLong test for correlated ROC curves. q is Benjamini–Hochberg over these nine contrasts. All p and q values here are descriptive (Section 6). Every comparison against the null excludes zero.

contrast	epoch	Δ AUC	95% CI	p	q
ENVELOPE − RANDOM	50	+0.0120	[+0.0068, +0.0173]	<0.0001	<0.0001
ENVELOPE − RANDOM	75	+0.0080	[+0.0028, +0.0132]	0.0025	0.0045
ENVELOPE − RANDOM	100	+0.0062	[+0.0010, +0.0113]	0.0199	0.0299
CENTROID − RANDOM	50	+0.0099	[+0.0051, +0.0149]	<0.0001	0.0001
CENTROID − RANDOM	75	+0.0113	[+0.0063, +0.0164]	<0.0001	<0.0001
CENTROID − RANDOM	100	+0.0109	[+0.0057, +0.0160]	<0.0001	<0.0001
CENTROID − ENVELOPE	50	-0.0020	[-0.0069, +0.0029]	0.4114	0.4114
CENTROID − ENVELOPE	75	+0.0033	[-0.0013, +0.0079]	0.1491	0.1677
CENTROID − ENVELOPE	100	+0.0047	[+0.0004, +0.0091]	0.0294	0.0378

Table 12: Full fp32 re-probe against the original fp16 result, same frozen encoder in each row. The largest difference is 2×10^{-4}, two to three orders of magnitude below the effects reported in Table 1.

policy	epoch	fp16 AUC	fp32 AUC	Δ	p
ENVELOPE	50	0.876064	0.876063	-0.000001	0.964
ENVELOPE	75	0.880307	0.880305	-0.000002	0.806
ENVELOPE	100	0.880743	0.880761	+0.000018	0.167
CENTROID	50	0.874030	0.874015	-0.000015	0.412
CENTROID	100	0.885485	0.885293	-0.000192	0.767
RANDOM	50	0.864097	0.864121	+0.000024	0.128
RANDOM	75	0.872302	0.872302	-0.000001	0.962
RANDOM	100	0.874581	0.874485	-0.000096	0.000

O Fine-tuned results

Table 13: Full fine-tuning, same test split. Reported separately from frozen probes and never pooled with them.

policy	head	test AUC
CENTROID	mean-pool	0.894668
CENTROID	cross-attention	0.893717
CENTROID	attentive	0.890100
RANDOM	attentive	0.887763
RANDOM	cross-attention	0.887178
RANDOM	mean-pool	0.886756

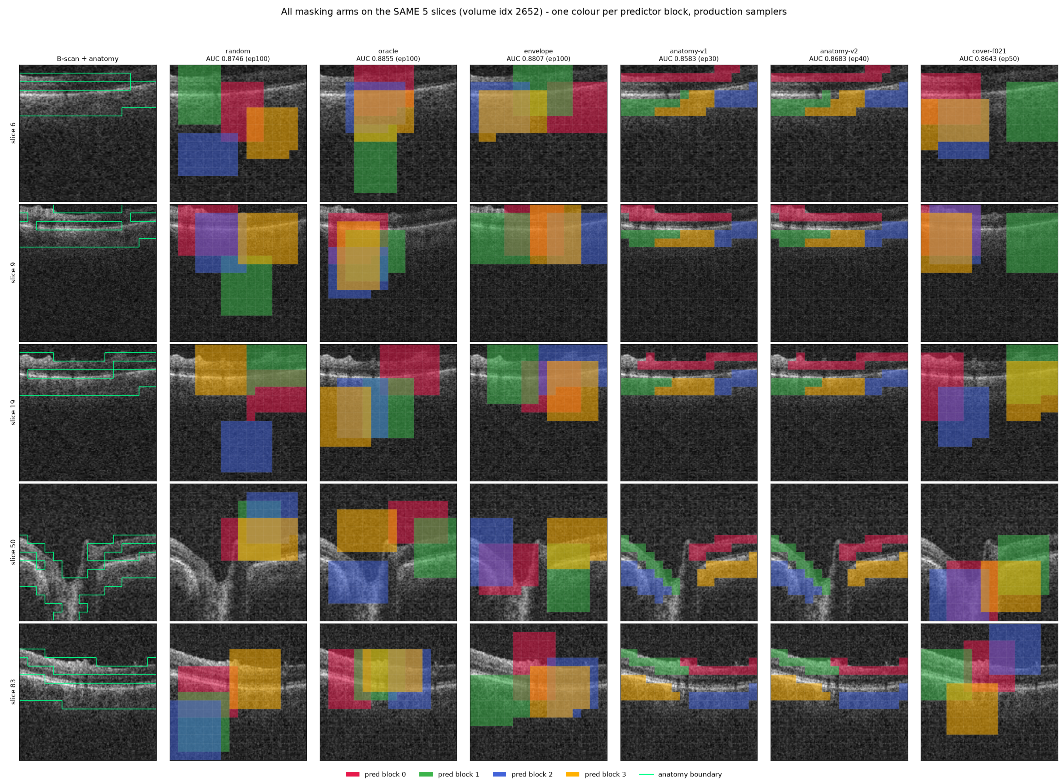

Figure 4: All six masking policies (columns) on five B-scans (rows) spanning one volume, drawn with the production samplers. Leftmost column is the unmasked B-scan. Each predictor target block has its own colour; the green contour is the anatomy reference. The random, envelope, and cover arms place axis-aligned rectangles; the anatomy arms grow cell-wise targets that follow the retinal band and therefore change shape from slice to slice.

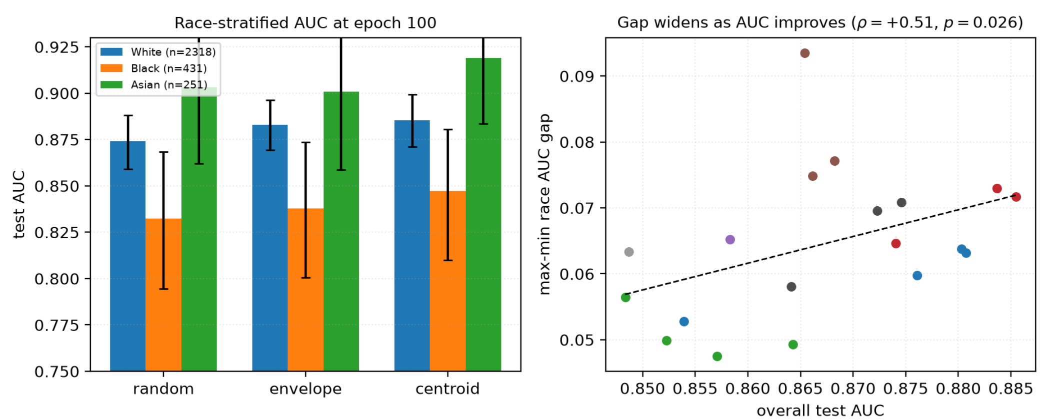

Figure 5: **Left:** race-stratified AUC at epoch 100 with bootstrap intervals. The black subgroup is lowest under every policy, but note that *every* group improves from RANDOM to CENTROID. **Right:** max–min race gap against aggregate AUC across 23 checkpoints. The fitted slope is positive and passes multiplicity correction at checkpoint level, but does not survive branch-level aggregation (Table 4), so we draw no trend conclusion from it.

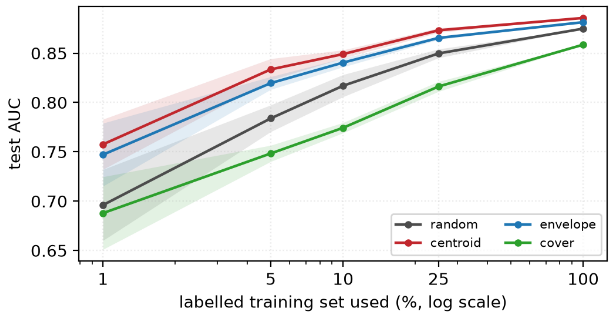

Figure 6: The policies converge as labels are added and separate as they are removed. Shading is one standard deviation over 5 label subsets, with every arm seeing identical subsets. At full supervision the four arms lie within 0.027 of one another; at 5% the spread is 0.085.

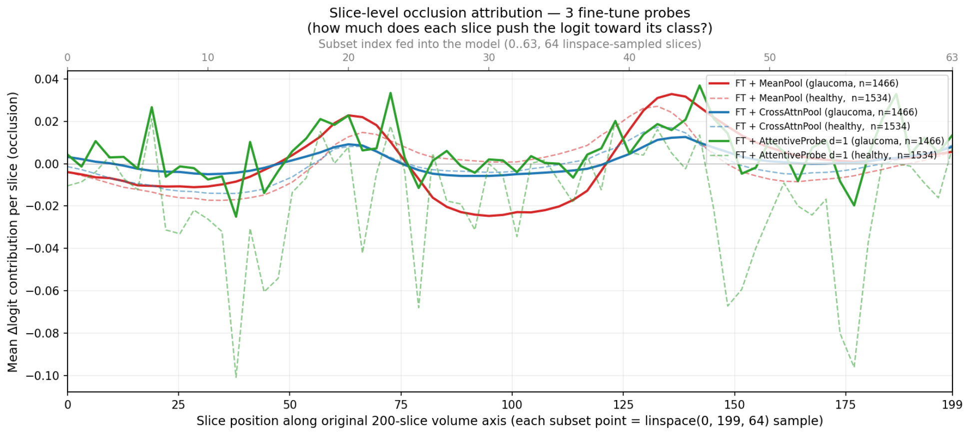

Figure 7: Single-slice occlusion attribution against native slice position, averaged over all 3000 test volumes, for all three fine-tuned probes. Despite spanning zero to 7.14M probe parameters they converge on the same structure (mean-pool vs cross-attention pool $r{=}0.94$). The y-axis is *signed* Δlogit: peaks are slices whose removal lowers the logit, the central dip is slices whose removal raises it, and only slices near zero are genuinely unused.

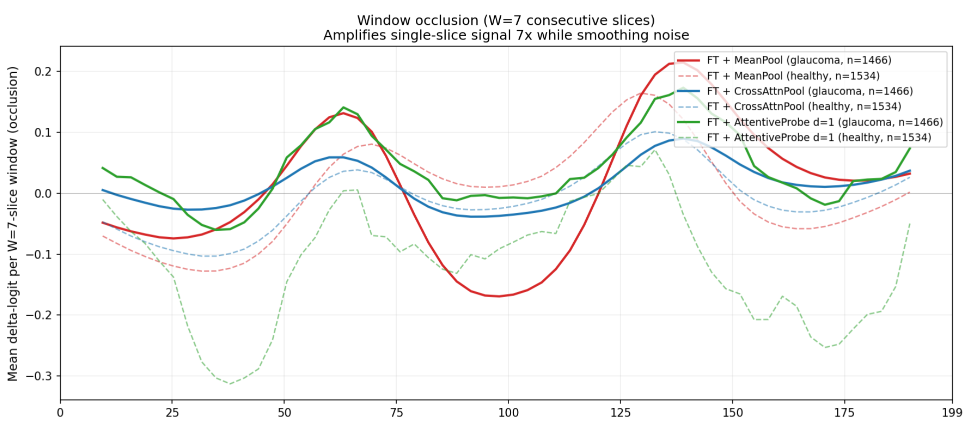

Figure 8: Window occlusion ($W{=}7$) amplifies the same signal roughly sevenfold and suppresses single-volume noise. For mean-pooled models, single-slice occlusion systematically underestimates attributed signal; the windowed variant recovers about $25\times$ more.

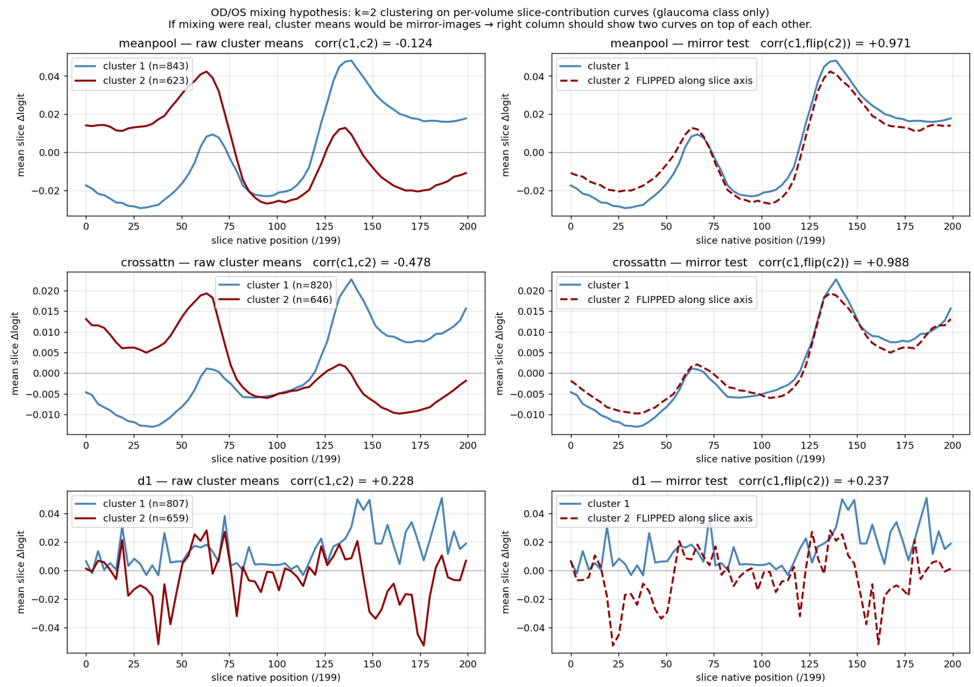

Figure 9: The two-peak structure is an OD/OS storage artefact. Clustering per-volume attribution curves into two groups returns near-perfect mirror images of one another, which is what laterality flipping would produce, rather than bilateral anatomy.

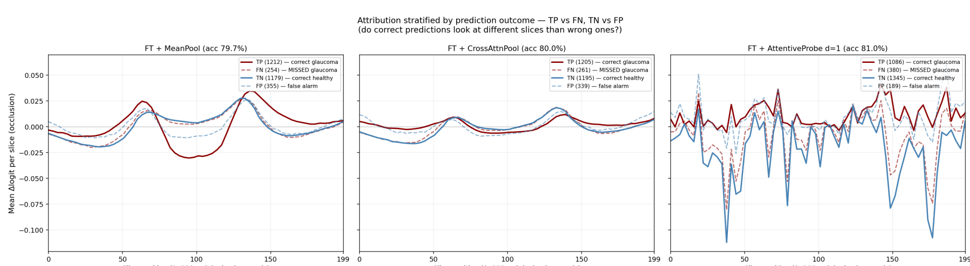

Figure 10: Attribution stratified by TP / FN / TN / FP. Errors reproduce the correct pattern at lower amplitude rather than selecting different anatomy.

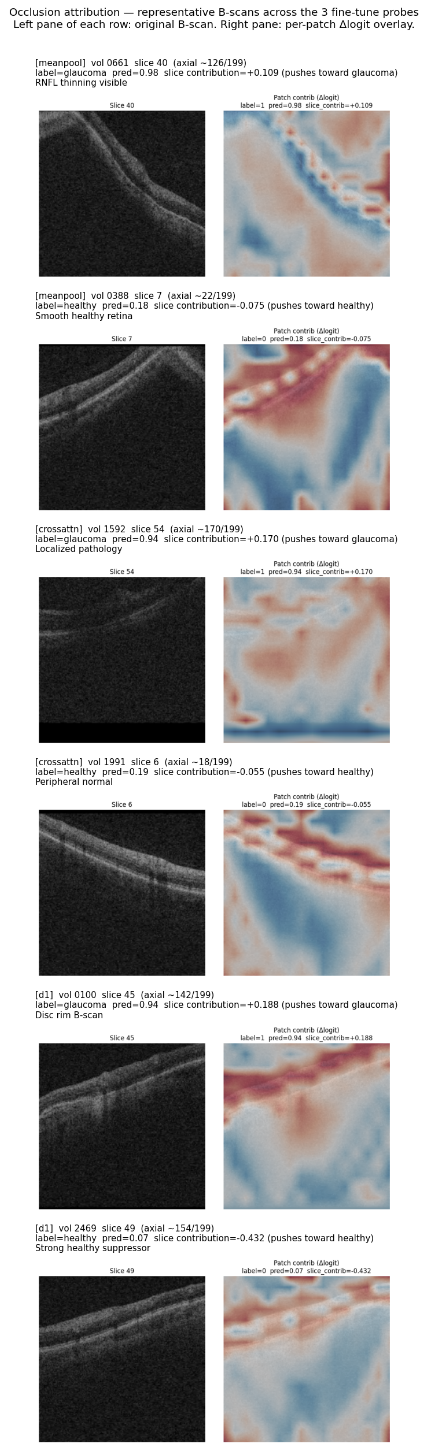

Figure 11: Per-patch occlusion heat maps on individual B-scans, drawn on a shared scale with the slice mean subtracted so that maps are comparable across volumes. Between 84% and 91% of patches have a 95% bootstrap interval excluding zero on the glaucoma class (B=500), so attribution is broadly distributed rather than concentrated in a few patches. Cross-probe agreement is strong at slice level (r=0.94) and moderate at patch level (per-volume $r \approx 0.35$–0.48).

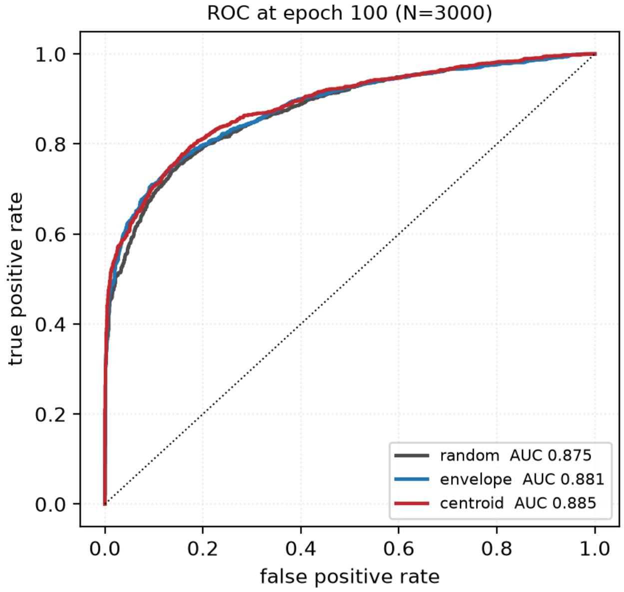

Figure 12: ROC curves at epoch 100 on the 3000 test volumes. The three arms plotted are nearly superimposed: the separation reported throughout this paper is a small, consistent shift in the high-specificity region rather than a difference visible in the curve as a whole. We include this so the effect size is not overstated by tables alone.

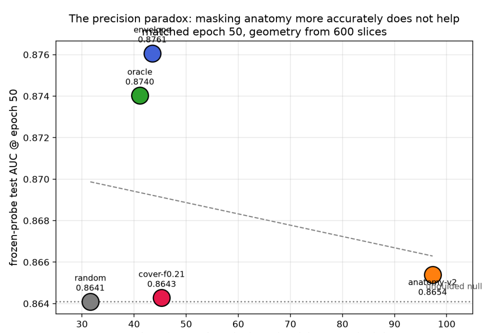

Figure 13: Downstream AUC at matched epoch 50 against mask purity, the fraction of masked patches lying on retinal tissue. The policy whose masks are 97% on-tissue does not separate from the null, while policies at 41–44% purity gain the most. The ordering does not support the intuition that more accurate anatomical masking yields better representations. With four to five arms this is descriptive, not inferential.

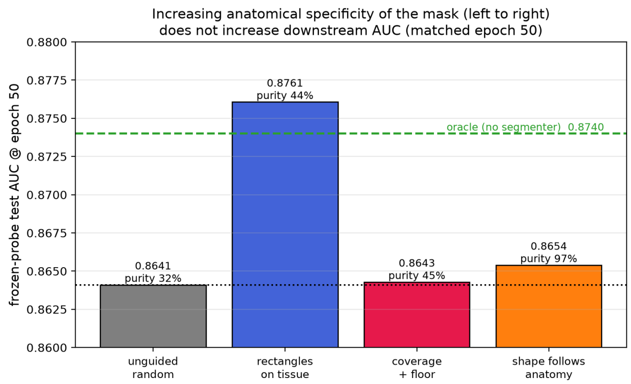

Figure 14: The matched-epoch-50 result arranged as a ladder of increasing anatomical specificity. Companion to Figure 13.

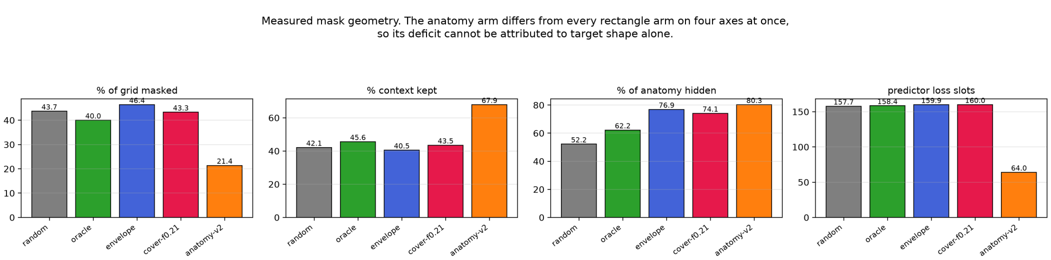

Figure 15: Measured mask geometry per policy, 600 held-out slices. The four rectangle policies are closely matched on every axis, so their comparison is near single-variable. The anatomy policy diverges on all four simultaneously — half the masking ratio, $1.6\times$ the context, $0.4\times$ the predictor loss slots — so its deficit cannot be attributed to target shape.